

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	30 July 2026
Team ID	Individual / Solo Project
Project Name	Automated Network Request Management in ServiceNow
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Maximum Marks	4 Marks

Technical Architecture / Workflow Flow

The diagram below illustrates the comprehensive architecture for the Automated Network Request solution. It maps the end-to-end data flow between the Employee Portal, the core ServiceNow Platform, and External IT/Network Management tools.

Employee	ServiceNow Platform (Cloud)	Manager / IT Admin
1. Employee Service Portal / Mobile App	→ Service Catalog & Request Workflow Flow Designer	
	2. ↓ Approval Engine Workflow Studio	Manager / InfoSec Approves / rejects in ServiceNow
	3. ↓ Risk & Policy Check CMDB / SecOps	→ Directory Service (AD / LDAP lookup) External API
	4. ↓ Fulfillment Orchestration Integration Hub	→ Network Mgmt Tool (Ansible / Firewall) External API → IT Admin Console Agent Workspace
approval req. ↔ decision	5. ↑ ServiceNow CMDB Cloud Database	
	6. completion & confirmation ← ← ← ← ← ← ←	

Table-1: Components & Technologies

S.No	Component	Description	Technology
1	Employee Portal	Interface for the employee to submit network access requests	Service Portal / Mobile App
2	Service Catalog Workflow	Trigger and request intake orchestrator	Flow Designer

S.No	Component	Description	Technology
3	Approval Engine	Routes requests to designated Managers / InfoSec personnel	Workflow Studio
4	Risk & Policy Check	Validates request against security baselines and active configurations	CMDB / SecOps
5	Directory Service	Looks up user identity and role-based permissions	AD / LDAP (External API)
6	Fulfillment Orchestration	Coordinates automated network deployment tasks	Integration Hub
7	Network Mgmt Tool	Executes the network/firewall configuration remotely	Ansible / Firewall API
8	IT Admin Console	Dashboard for IT staff to monitor request fulfillment	Agent Workspace
9	Cloud Database	Stores the authoritative network topology and Configuration Items	ServiceNow CMDB

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Distributed Orchestration	Flow Designer seamlessly coordinates internal approvals and external API actions	Integration Hub
2	Automated Compliance	Pre-deployment checks ensure requests meet InfoSec policies	SecOps Integration
3	Scalable Architecture	Multi-instance SaaS architecture supports high-volume request processing	ServiceNow Now Platform
4	Dynamic CMDB Updates	Automatically updates CI relationships upon successful network configuration	GlideRecord / REST APIs
5	Real-time Tracking	Employees receive immediate status updates via the portal upon completion	ServiceNow Notifications